

NGUYEN BA LONG

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Summary

Final-year Computer Science student at UIT specializing in Computer Vision, Deep Learning, and Document AI/OCR. Experienced in building end-to-end multimodal pipelines—from image preprocessing and custom architecture design to high-throughput OCR extraction (PaddleOCR) and CPU/GPU inference optimization. Seeking a Computer Vision/OCR Intern position to apply research-driven solutions to real-world intelligent document processing challenges.

Education

University of Information Technology (UIT) - VNUHCM

Sep. 2023 - Jun. 2027 (Expected)

Bachelor of Science in Computer Science

Ho Chi Minh City, Vietnam

- **GPA:** 8.27/10.0
- **Relevant Coursework:** Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Data Mining, Big Data.

Technical Skills

Languages: Python, C/C++, SQL, JavaScript

AI/ML & Deep Learning: PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face Transformers

Generative AI & RAG: LangChain, LangGraph, Prompt Engineering, Vector Search, Hybrid Retrieval, Reranking

Computer Vision & Multimodal: OpenCV, PaddleOCR, Image Segmentation, VLM, Dataset Annotation

Data Processing: ETL Pipelines, Data Cleaning, Feature Engineering, Distributed Processing

Backend & Databases: FastAPI, Flask, RESTful APIs, PostgreSQL, Vector & Search Databases, Docker, Git

Projects

Temporal RAG for UIT Course Video Q&A | *LangGraph, Qwen2-VL, Whisper, FastAPI, Chroma*

[GitHub](#)

- Built an agentic RAG pipeline with LangGraph to answer questions across 55 lecture videos, integrating query guardrails and HyDE transformation to improve search context.
- Extracted slide text using Qwen2-VL and audio transcripts using Whisper-large-v3, synchronizing multimodal content into structured chunks with exact video timestamps.
- Implemented a hybrid retrieval engine combining BM25 keyword search and Chroma vector search (multilingual-e5-large) via Reciprocal Rank Fusion (RRF, $k = 60$) and BGE-Reranker.
- Developed an asynchronous FastAPI backend to return answers with clickable video timestamps; benchmarked the system on 90 test queries via RAGAS, reaching 0.96 Faithfulness and 0.89 Answer Relevancy.

AIC Multimodal Video Retrieval System | *PaddleOCR, SigLIP2, YOLOv11, Milvus, Elasticsearch, Redis*

[GitHub](#)

- Engineered a multimodal ingestion pipeline for 726 videos across 5 modalities (SigLIP2, Whisper, PaddleOCR, Florence-2, YOLOv11); handled dataset preparation and frame annotation.
- Built an OCR & ASR extraction pipeline using PaddleOCR (PP-OCRv5) and Whisper-large-v3, applying OpenCV image preprocessing and Vietnamese prompt engineering to extract on-screen text and speech transcripts.
- Developed a hybrid multi-stream retrieval engine combining Milvus dense vector search (SigLIP2) and Elasticsearch BM25 sparse search via Reciprocal Rank Fusion (RRF, $k = 60$), fine-tuned with Cross-Encoder (BGE-Reranker).
- Integrated query decomposition (Gemini API / rule-based fallback), interactive Rocchio Relevance Feedback, and Redis caching to achieve sub-second search latency across KIS, Visual Q&A, and TRAKE queries.

HTS-MultiResUNet: Dual-Attention Hybrid CNN-Transformer | *TensorFlow, Keras, Scikit-learn, OpenCV* [GitHub](#)

- Engineered a hybrid CNN-Transformer architecture for multimodal biomedical image segmentation (Fluorescence, Electron, Dermoscopy, Endoscopy), coupling Squeeze-and-Excitation (SE) channel attention with a Multi-Head Self-Attention (MHSA) bottleneck to capture long-range contextual dependencies.
- Constructed a robust training pipeline using 5-fold cross-validation (Scikit-learn), strict image normalization, and gradient stabilization techniques to handle severe spatial class imbalances across 4 distinct datasets; evaluated model performance using IoU and Dice metrics.
- Formulated an Edge-Aware Focal Tversky Loss (EA-FTL) with a discrete Sobel operator to preserve fine anatomical boundaries, outperforming the baseline MultiResUNet by 5.83% in Jaccard Index (IoU) and achieving superior Dice scores.

Languages

Languages: Vietnamese (Native)

English: Professional Reading Proficiency; TOEIC Speaking & Writing 250/400.